MALLA REDDY COLLEGE OF ENGINEERING & TECHNOLOGY

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DEPARTMENT OF AERONAUTICAL ENGINEERING

APP DEVELOPMENT 1

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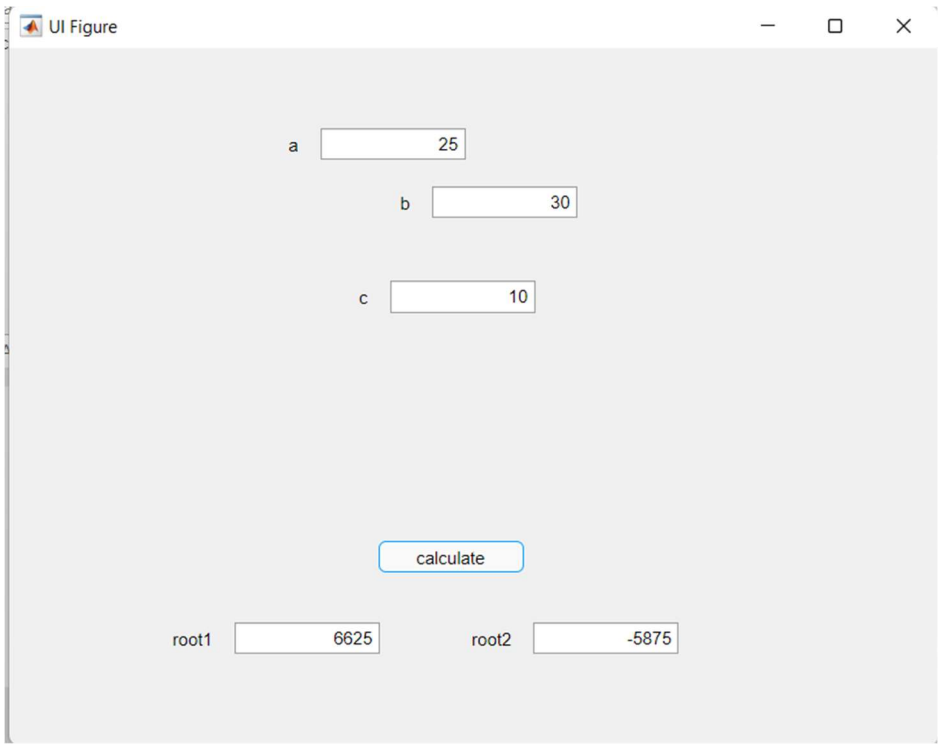
CODE VIEW:

```
classdef qadeqn < matlab.apps.AppBase


    % Properties that correspond to app components
    properties (Access = public)
        UIFigure             matlab.ui.Figure
        calculateButton      matlab.ui.control.Button
        aEditFieldLabel      matlab.ui.control.Label
        aEditField           matlab.ui.control.NumericEditField
        bEditFieldLabel      matlab.ui.control.Label
        bEditField           matlab.ui.control.NumericEditField
        root1EditFieldLabel  matlab.ui.control.Label
        root1EditField       matlab.ui.control.NumericEditField
        root2EditFieldLabel  matlab.ui.control.Label
        root2EditField       matlab.ui.control.NumericEditField
        cEditFieldLabel      matlab.ui.control.Label
        cEditField           matlab.ui.control.NumericEditField
    end


    % Callbacks that handle component events
    methods (Access = private)


        % Button pushed function: calculateButton
        function calculateButtonPushed(app, event)
            a=app.aEditField.Value;
            b=app.bEditField.Value;
            c=app.cEditField.Value;
            root1=(b+(4*a*c)^1/2)/2*a;
            root2=(b-(4*a*c)^1/2)/2*a;
            app.root1EditField.Value=root1;
            app.root2EditField.Value=root2;

        end
    end


    % Component initialization
    methods (Access = private)


        % Create UIFigure and components
        function createComponents(app)


            % Create UIFigure and hide until all components are created
            app.UIFigure = uifigure('Visible', 'off');
            app.UIFigure.Position = [100 100 640 480];
            app.UIFigure.Name = 'UI Figure';


            % Create calculateButton
```

```
            app.calculateButton = uibutton(app.UIFigure, 'push');
            app.calculateButton.ButtonPushedFcn = createCallbackFcn(app,
@calculateButtonPushed, true);
            app.calculateButton.Position = [255 120 100 22];
            app.calculateButton.Text = 'calculate';


            % Create aEditFieldLabel
            app.aEditFieldLabel = uilabel(app.UIFigure);
            app.aEditFieldLabel.HorizontalAlignment = 'right';
            app.aEditFieldLabel.Position = [175 404 25 22];
            app.aEditFieldLabel.Text = 'a';


            % Create aEditField
            app.aEditField = uieditfield(app.UIFigure, 'numeric');
            app.aEditField.Position = [215 404 100 22];


            % Create bEditFieldLabel
            app.bEditFieldLabel = uilabel(app.UIFigure);
            app.bEditFieldLabel.HorizontalAlignment = 'right';
            app.bEditFieldLabel.Position = [252 364 25 22];
            app.bEditFieldLabel.Text = 'b';


            % Create bEditField
            app.bEditField = uieditfield(app.UIFigure, 'numeric');
            app.bEditField.Position = [292 364 100 22];


            % Create root1EditFieldLabel
            app.root1EditFieldLabel = uilabel(app.UIFigure);
            app.root1EditFieldLabel.HorizontalAlignment = 'right';
            app.root1EditFieldLabel.Position = [108 64 33 22];
            app.root1EditFieldLabel.Text = 'root1';


            % Create root1EditField
            app.root1EditField = uieditfield(app.UIFigure, 'numeric');
            app.root1EditField.Position = [156 64 100 22];


            % Create root2EditFieldLabel
            app.root2EditFieldLabel = uilabel(app.UIFigure);
            app.root2EditFieldLabel.HorizontalAlignment = 'right';
            app.root2EditFieldLabel.Position = [314 64 33 22];
            app.root2EditFieldLabel.Text = 'root2';


            % Create root2EditField
            app.root2EditField = uieditfield(app.UIFigure, 'numeric');
            app.root2EditField.Position = [362 64 100 22];
```

```
            % Create cEditFieldLabel
            app.cEditFieldLabel = uilabel(app.UIFigure);
            app.cEditFieldLabel.HorizontalAlignment = 'right';
            app.cEditFieldLabel.Position = [223 299 25 22];
            app.cEditFieldLabel.Text = 'c';

            % Create cEditField
            app.cEditField = uieditfield(app.UIFigure, 'numeric');
            app.cEditField.Position = [263 299 100 22];

            % Show the figure after all components are created
            app.UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = qadeqn

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.UIFigure)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.UIFigure)
        end
    end
end
```

RESULT :-